Machine Learning Assignment

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Programme: B.Tech Computer Science and Engineering

Summary

Table 1. Summary of the main results.

Part	Dataset and scale	Models and methods	Primary empirical finding
1. Classification	EuroSAT RGB, 27,000 images, 10 classes (64×64×3 pixels)	KNN (5-fold CV), Decision tree (Gini), SVM (RBF kernel), Deep CNN (ResNet-18)	ResNet-18 dominates all models with 94.30% test accuracy and 0.9977 macro ROC-AUC. The RBF SVM is a strong second (72.70%, AUC 0.9462). KNN with cosine distance reaches 53.65% (AUC 0.8776), beating the decision tree (32.89%).
2. Regression	Metro Traffic Volume, 48,204 hourly records, 23 engineered features	Linear (batch GD), Polynomial (degree 1–5), Ridge and lasso	Sensible linear and polynomial models reach test $R^2 \approx 0.697$–0.699 (RMSE ≈ 1,080 veh/h). Degree 5 is severely ill-conditioned (condition number 1.8×10^{27}). Ridge $\alpha = 100$ and lasso $\alpha = 10$ chosen by 5-fold time-series CV.
3A. Clustering	Gas turbine telemetry, 36,726 hourly records, 11 sensor variables	K-means from scratch (Lloyd's algorithm, 10 random restarts)	WCSS evaluated for K = 2 to 6; the elbow curve bends gradually. K = 3 is adopted as an informative operational partition into low, medium and high turbine load.
3B. PCA	Gas turbine telemetry, 11 standardized variables	PCA from scratch (eigen-decomposition of $C = X^TX/(N-1)$)	PC1 captures 48.00% and PC2 19.90% (cumulative 67.90% for 2D, 77.35% for 3D). PC1 is a turbine-load axis dominated by CDP, GTEP, TEY and TIT.

1. Overview and Experimental Setup

To satisfy the core objectives of the machine learning curriculum, three standard benchmark datasets were analyzed. Following the rubric, all tasks use strict training–test separation with no data leakage. Preprocessing parameters (scalers, feature extractors and tuning bounds) were fitted exclusively on training partitions.

Part 1 (Classification) benchmarks four multiclass models on the EuroSAT RGB satellite dataset (27,000 images, 10 land-cover classes, 64×64×3): K-nearest neighbours with 5-fold cross-validation, a decision tree with complexity control, a support vector machine with an RBF kernel, and an adapted deep convolutional neural network (ResNet-18) for the 5-mark bonus. Part 2 (Regression) investigates batch gradient-descent linear regression, polynomial regression (degrees 1 to 5), ridge and lasso regularization on 48,204 chronological records of Interstate 94 traffic volume. Part 3 (Unsupervised Learning) implements K-means clustering with 10 restarts and principal component analysis from scratch in NumPy on 36,726 hourly gas-turbine sensor observations. Preprocessing is fitted on training data only, and the test data are used only for the final evaluation.

Table 2. Overview of benchmark datasets and task scopes.

Part	Dataset source	Records / samples	Task description and scope
Part 1	EuroSAT RGB (Sentinel-2 satellites)	27,000 images (64×64×3 RGB)	Multiclass land-cover classification into 10 categories. Traditional ML (HOG features) versus deep CNN (raw spatial tensors).
Part 2	Metro Interstate Traffic Volume (UCI 492)	48,204 hourly records	Continuous traffic volume prediction (veh/h). Batch gradient descent, polynomial degrees 1–5, ridge and lasso.
Part 3	Gas Turbine Emissions (UCI 551)	36,726 clean records (11 sensors)	Unsupervised operating-state clustering via K-means and dimensionality reduction via PCA, both coded from scratch in NumPy.

2. Part 1: Multiclass Classification (EuroSAT Satellite Imagery)

2.1 Dataset exploration and stratified 80/20 partitioning

The EuroSAT RGB dataset contains 27,000 georeferenced satellite images acquired by the European Space Agency's Sentinel-2 platform covering cities across 34 European countries. Each image has a

spatial resolution of 64×64 pixels across 3 RGB channels (12,288 raw values). The dataset spans 10 land-cover classes: AnnualCrop (3,000), Forest (3,000), HerbaceousVegetation (3,000), Highway (2,500), Industrial (2,500), Pasture (2,000), PermanentCrop (2,500), Residential (3,000), River (2,500) and SeaLake (3,000). The classes are well balanced (Figure 1).

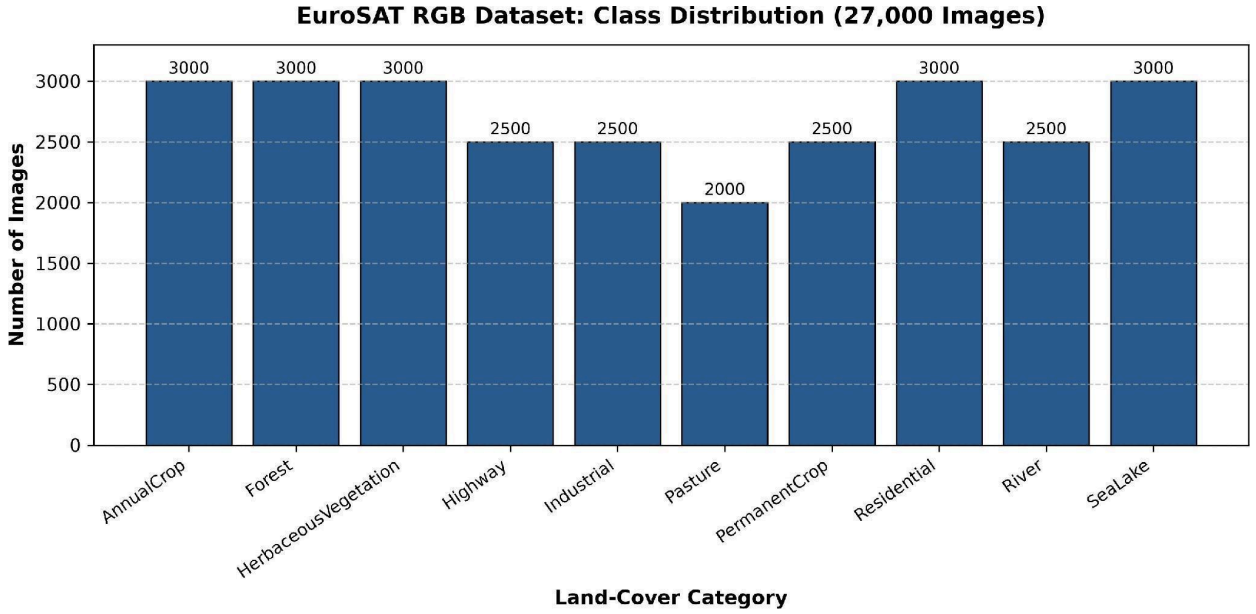


Figure 1. EuroSAT class distribution (27,000 images).

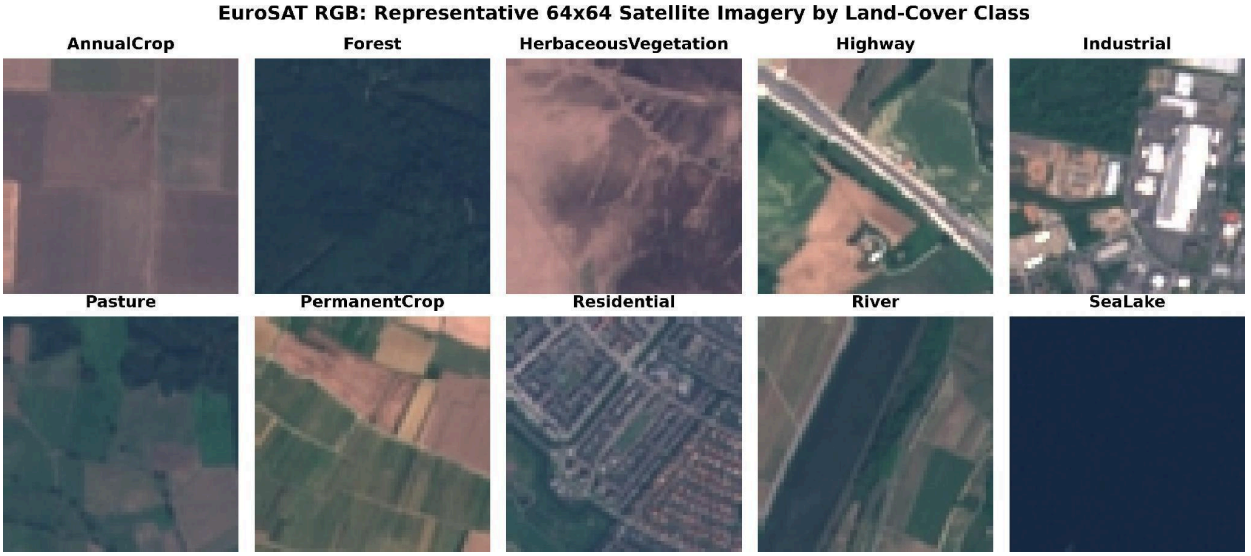


Figure 2. Representative 64×64 images across all 10 classes.

An exact 80% training / 20% testing split was generated using stratified sampling (random seed 42), producing 21,600 training images and 5,400 held-out test images. The partition indices were written to disk (eurosat_split.json) so that all four classifiers were evaluated on the identical 5,400 test images. For neural network training, a 10% validation split (2,700 images) was carved from the training partition for checkpoint selection (best validation loss; early stopping did not trigger), leaving 18,900 pure training images.

2.2 Feature extraction for traditional ML (HOG)

KNN, the decision tree and the SVM require 1D numerical vectors. Histogram of Oriented Gradients (HOG) was used to compress spatial edge and gradient-orientation distributions. With 9 orientation bins, 8×8 pixels per cell and 2×2 cells per block evaluated on the channel with the largest gradient at each pixel, each 64×64 image produces one 1,764-dimensional descriptor (colour values are not encoded). HOG extraction was computed across 16 CPU cores and cached to disk. StandardScaler was included inside the scikit-learn Pipeline objects for KNN and SVM, so test-data statistics never contaminated the training normalization.

2.3 Model training and cross-validation

KNN, tree and SVM hyper-parameters were tuned on a stratified 5,000-image training subset (about 4,000 images per fold). The chosen models were then refitted on all 21,600 images, which is why test accuracy exceeds CV accuracy.

• K-nearest neighbours. Metric in {euclidean, cosine}, K in {1, 3, 5, 7, 9, 11, 15, 21} with uniform and distance weighting were evaluated using 5-fold stratified CV. The optimal setting was K = 15 with distance weighting and cosine distance (5-fold CV accuracy 52.28%).

• Decision tree. Depth limits {10, 15, 20, 25, None}, minimum split samples {2, 5, 10} and Gini/entropy criteria were evaluated using 5-fold stratified CV. The optimal configuration was Gini impurity with maximum depth 10 and minimum samples split 2 (CV accuracy 29.22%, 682 leaf nodes).

• Support vector machine (RBF kernel). A grid search over C in {0.1, 1.0, 10.0} and γ in {'scale', 'auto'} was performed with 3-fold stratified CV. After StandardScaler, γ='scale' and 'auto' are identical (1/1764), so only C was effectively tuned. The optimal configuration was C = 10.0, γ = 'scale' (CV accuracy 65.80%). The final model used 17,310 support vectors (80.1% of training data), trained on all 21,600 samples. Decision scores were converted to probabilities via softmax for continuous ROC-AUC.

• Deep CNN (ResNet-18). ResNet-18 was trained from scratch (no pre-trained weights) for 6 epochs. The architecture was adapted for 64×64 images by replacing the initial 7×7 conv with 3×3 conv (stride 1, pad 1), setting initial max-pooling to identity, keeping all 4 residual stages (64→128→256→512 channels), and projecting global-average-pooled features to 10 class logits. Training used AdamW (lr 0.001, decay 10−4), cosine annealing, and random flips on an NVIDIA RTX 3060 GPU.

Table 3. Performance comparison on the untouched EuroSAT test set (5,400 images).

Classifier	**Acc.**	**Macro prec.**	**Macro recall**	**Macro F1**	**Macro ROC-AUC**	**Train acc.**	**Gen. gap**	**Train time**	**Test pred. time**
Deep CNN (ResNet-18)	94.30%	0.9411	0.9417	0.9412	0.9977	94.47%	0.17%	339.4 s	8.83 s
SVM (RBF kernel)	72.70%	0.7226	0.7157	0.7181	0.9462	100.0%	27.30%	8,836.8 s	141.77 s
KNN (with 5-fold CV)	53.65%	0.5798	0.5096	0.4827	0.8776	100.0%	46.35%	65.3 s	1.94 s
Decision tree	32.89%	0.3289	0.3165	0.3183	0.7053	51.89%	19.00%	463.1 s	0.02 s

Note: Train time covers hyper-parameter search plus final fit for KNN, tree and SVM, and 6 epochs with validation for ResNet-18. ResNet train accuracy is the running accuracy with augmentation on.

2.4 Confusion matrix analysis

Figure 3 shows the 4-panel normalized confusion matrices of all classifiers. The diagonal entries are the recall per category, while the off-diagonals reveal systematic confusions.

- KNN (cosine, K = 15): Recall is 89.2% for AnnualCrop, 82.5% for Forest and 93.5% for SeaLake. It struggles severely on Pasture (3.5%) and PermanentCrop (5.0%). AnnualCrop is heavily over-predicted, accounting for 1,772 of the 5,400 predictions (32.8%), and absorbs 57.6% of PermanentCrop, 52.4% of Industrial, and 34.8% of River.

- Decision tree: Suffers widespread misclassification across visually similar vegetation and textures. The largest off-diagonal confusions are Pasture→HerbaceousVegetation (23.8%), Industrial→HerbaceousVegetation (19.6%), Forest→HerbaceousVegetation (19.2%) and AnnualCrop→River (19.0%).

- SVM (RBF kernel): Resolves bulk land cover effectively. SeaLake recall is 93.7% and Forest is 87.2%. The weakest classes are PermanentCrop (45.0%) and Pasture (57.0%). The largest confusions are Pasture→HerbaceousVegetation (15.8%), Industrial→PermanentCrop (14.4%) and PermanentCrop→HerbaceousVegetation (13.2%).

- Deep CNN (ResNet-18): Eight of ten classes achieve recall above 90% (Forest 99.5%, SeaLake 98.3%, Residential 98.3%). The lowest recalls are PermanentCrop (89.2%) and River (88.2%), demonstrating robust hierarchical feature extraction.

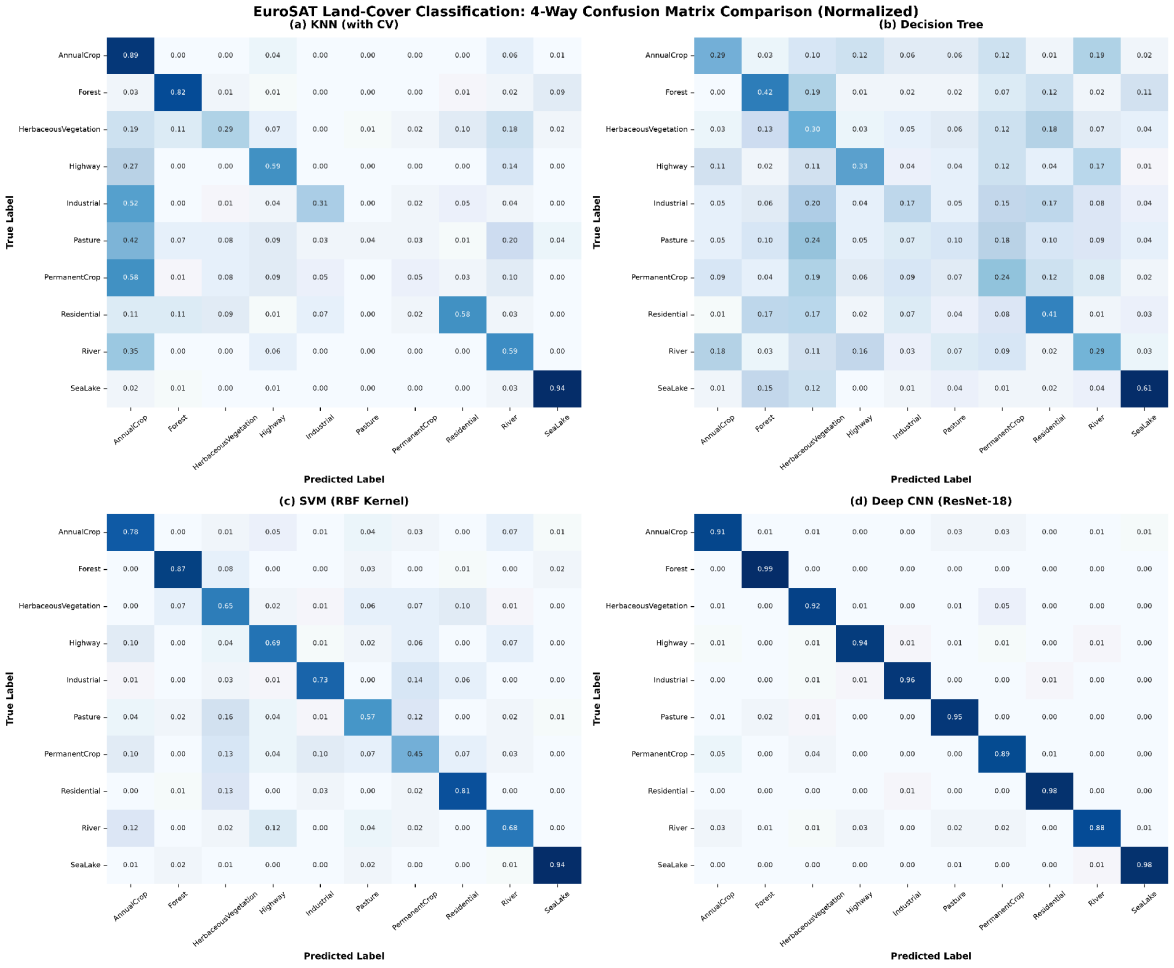


Figure 3. Normalized 4-way confusion matrix comparison on the EuroSAT test set (5,400 images).

2.5 ROC analysis and best classifier

Figure 4 presents the comparative macro-average one-vs-rest ROC curves. Continuous probability scores were used for all four models (softmax logits for ResNet-18, softmax of decision margins for the SVM, frequency proportions for the decision tree, and neighbour voting weights for KNN).

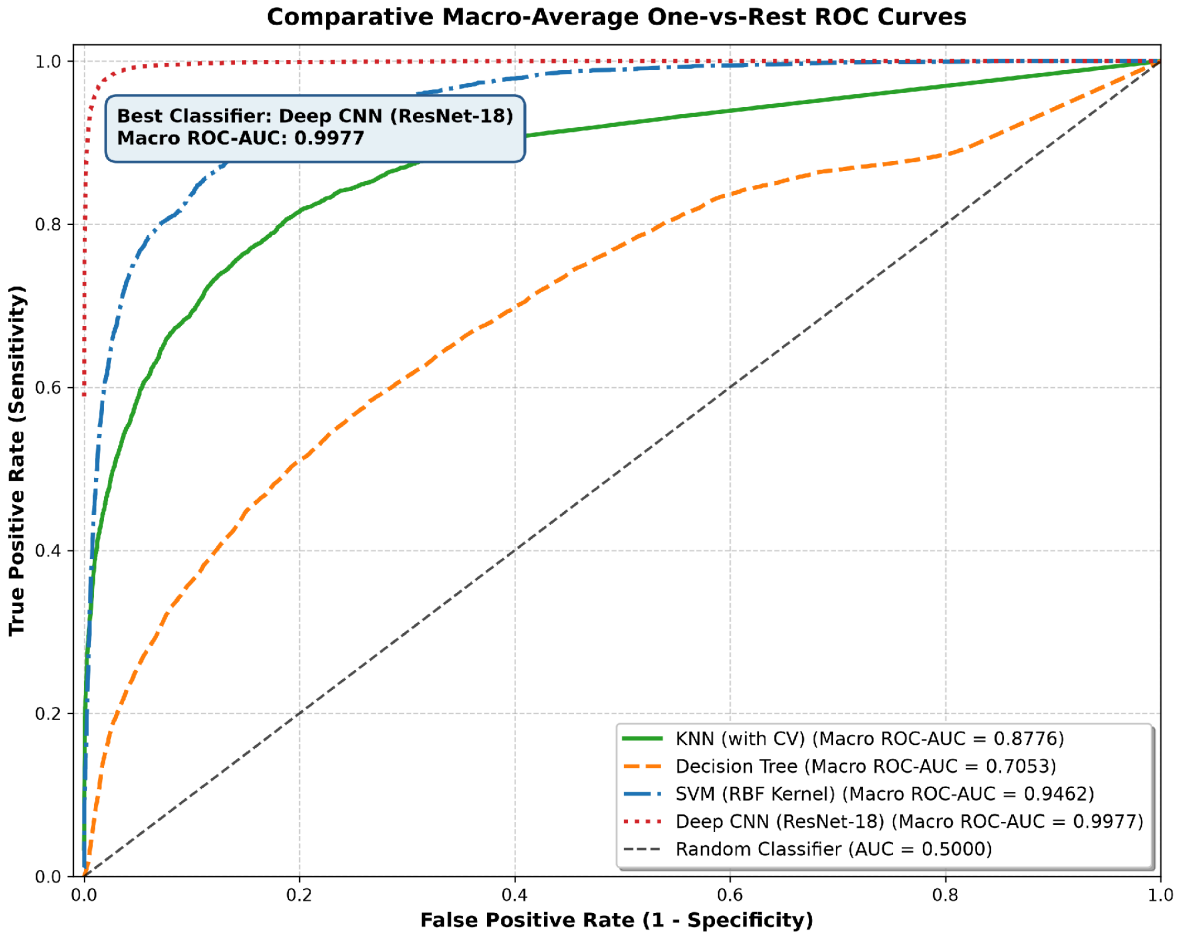


Figure 4. Comparative macro one-vs-rest ROC curves on the EuroSAT test set.

Best classifier based on ROC analysis: the deep CNN (ResNet-18). Its macro ROC-AUC reaches 0.9977, showing near-perfect ranking across all decision thresholds; its ROC curve rises almost vertically along the y-axis, reaching a true positive rate above 0.95 at false positive rates below 0.01. The RBF SVM is a competitive second (ROC-AUC = 0.9462), followed by KNN (0.8776) and the decision tree (0.7053). ResNet-18 lies above all three traditional models along the ROC trajectory.

2.6 Overfitting and model complexity

The four models show very different complexity dynamics and generalization gaps (train accuracy minus test accuracy).

• KNN (K = 15, cosine, distance-weighted): 46.35% generalization gap (100.0% train vs 53.65% test, 0.8776 ROC-AUC). Euclidean KNN collapsed onto Forest and SeaLake (66% of predictions); CV selected cosine distance, and AnnualCrop is now over-predicted (1,772 of 5,400 predictions), absorbing substantial PermanentCrop and Industrial samples. Test accuracy improves to 53.65%, beating the decision tree while remaining constrained by the lack of hierarchical spatial representations.

• Decision tree (depth 10, Gini): optimal bias–variance trade-off at depth 10. Test accuracy by depth is 22.83%, 27.39%, 32.89%, 31.54%, 31.59% and 30.31% at depths 3, 5, 10, 15, 20 and unlimited, while train accuracy rises monotonically from 24.7% to 100.0%. Decision trees build orthogonal, axis-aligned splits that fragment continuous gradient vectors inefficiently.

- **SVM (RBF, C = 10, γ = 'scale'): high-capacity regularized margin (27.30% gap).** With C = 10.0 the penalty on misclassification is high, producing 17,310 support vectors (80.1% of the training data). The model attains 100% training accuracy and a test accuracy of 72.70%. The 27.30% gap reflects boundary memorization of noisy vegetation textures, yet the regularized margin prevents catastrophic variance.

- **Deep CNN (ResNet-18): superior inductive bias and minimal overfitting (0.17% gap).** ResNet-18 has 11.2 million parameters, far more than the traditional models, yet shows almost no generalization gap (94.47% train versus 94.30% test). Convolutional parameter sharing, spatial pooling and translation equivariance provide a suitable inductive bias for 2D imagery. Residual skip connections ease gradient flow, while batch normalization and data augmentation regularize the representation.

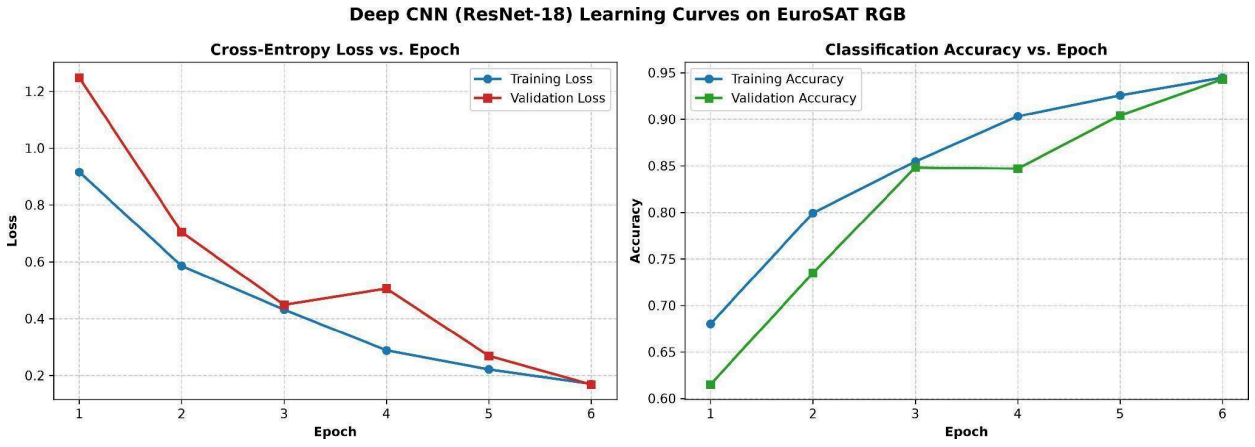


Figure 5. ResNet-18 learning curves (loss and accuracy).

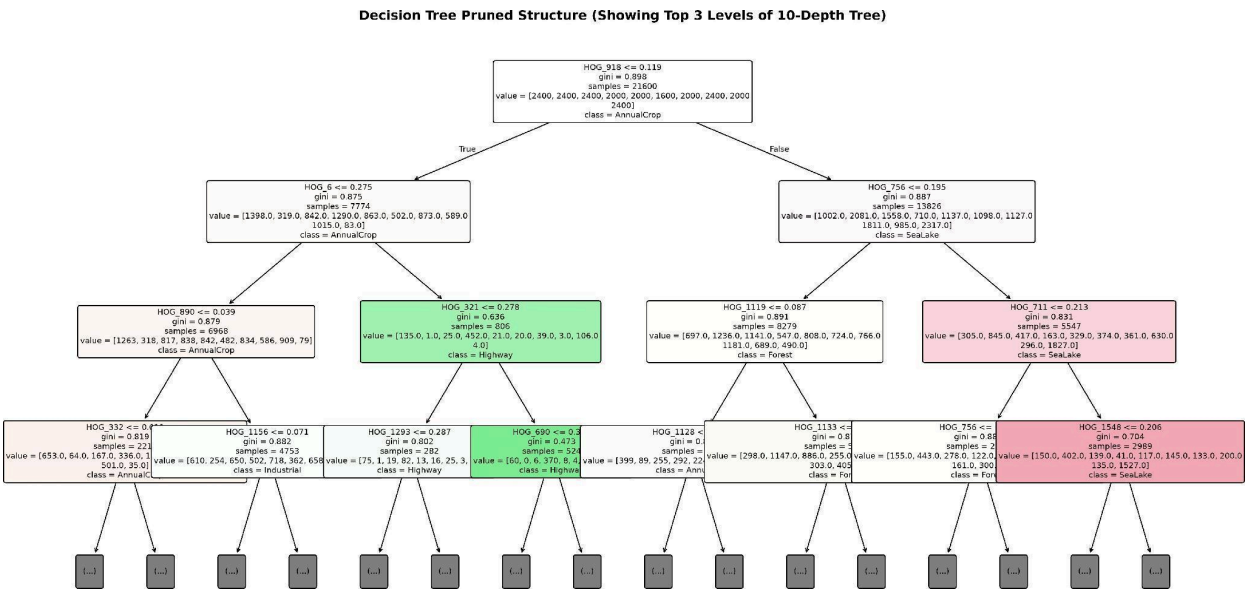


Figure 6. Decision tree, top levels (pruned visualization).

2.7 Summary of Part 1

All rubric criteria are covered: stratified 80/20 partitioning on EuroSAT, 5-fold cross-validation for KNN, a complexity-controlled decision tree, an RBF SVM with probability scores, and a deep ResNet-18 (bonus). ResNet-18 achieves the best accuracy (94.30%), F1-score (0.9412) and ROC-AUC (0.9977).

3. Part 2: Regression (Interstate Traffic Volume Prediction)

3.1 Data and preprocessing

The dataset contains 48,204 hourly records of westbound Interstate 94 traffic volume between Minneapolis and St. Paul, Minnesota, with weather and calendar variables. Traffic volume is a continuous temporal process with strong autocorrelation, so random shuffling would inject future diurnal cycles into past intervals and cause temporal leakage. The data were therefore sorted chronologically: the first 80% (38,563 records; October 2012 to 1 November 2017) form the training set and the final 20% (9,641 records; 1 November 2017 to 30 September 2018) form the held-out test set. Ten sensor-failure records with a temperature of 0 K were replaced with the training monthly median. A physically impossible rainfall reading of 9,831.3 mm/h was capped at the training 99.99th percentile (29.14 mm). The model uses 23 features: standardized continuous variables (temperature, rainfall, snowfall, cloud cover), sine/cosine cyclic encodings of hour, day of week and month, binary holiday and weekend flags, and 11 one-hot weather-condition indicators.

3.2 Linear regression with batch gradient descent

The hypothesis is $\hat{y} = Xw + b$, optimized on the cost $J(w, b) = (1/2N)\,\Sigma\,(\hat{y}_i - y_i)^2$ with the updates $w \leftarrow w - \eta\,(1/N)\,X^T(\hat{y} - y)$ and $b \leftarrow b - \eta\,(1/N)\,\Sigma(\hat{y}_i - y_i)$, using learning rate $\eta = 0.05$ over 1,000 iterations. Figure 7 shows the monotonic cost reduction. The optimization stopped at the 1,000-iteration budget rather than meeting the 10^{-6} tolerance (the last change in cost was 0.71). Compared with the closed-form ordinary least squares (OLS) solution, the gradient-descent test MSE is 1,171,125 (RMSE 1,082.2 veh/h, R^2 = 0.6975) versus 1,173,326 for OLS (RMSE 1,083.2 veh/h, R^2 = 0.6969). The predictions differ by only 15.8 veh/h (0.49% of mean traffic). The slight test advantage of gradient descent is attributed to early stopping acting as mild implicit shrinkage (weight norm 2,434 versus 2,535 for OLS).

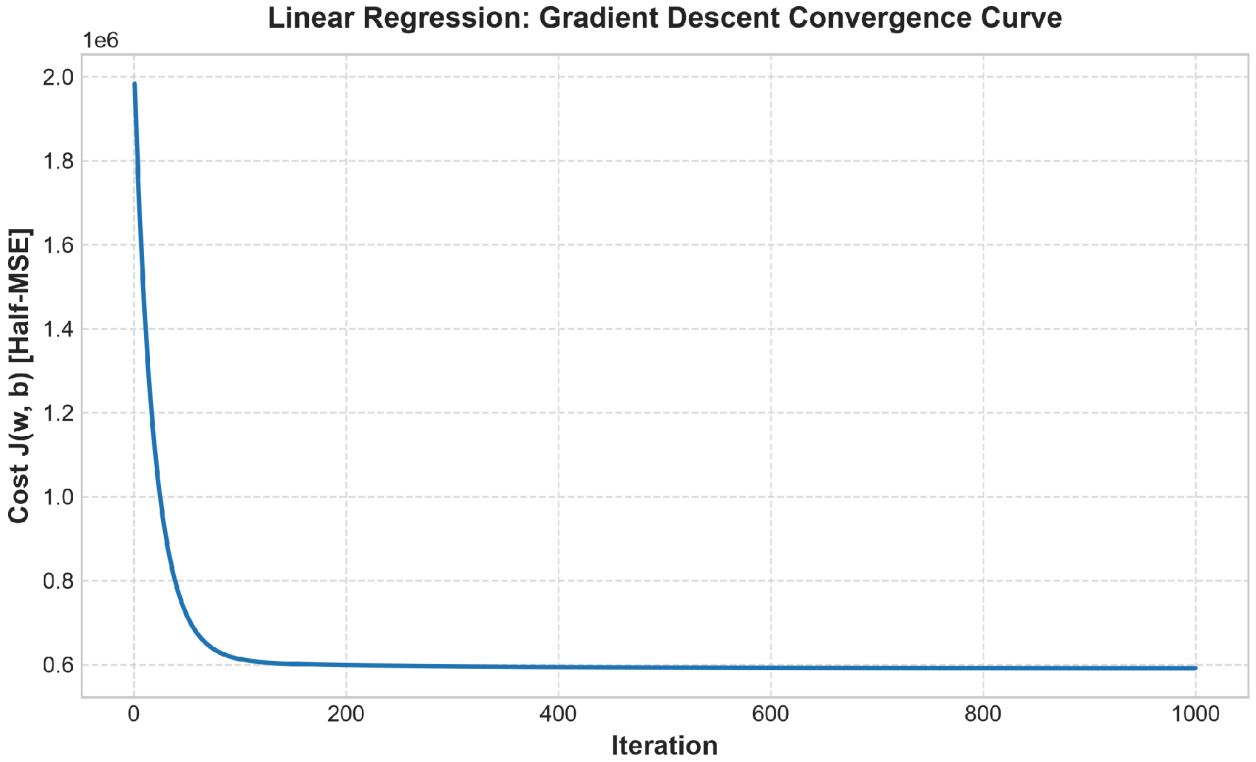


Figure 7. Batch gradient descent cost against iteration over 1,000 iterations.

3.3 Polynomial regression and numerical conditioning

Polynomial features of degree 1 to 5 were generated for the four standardized continuous variables only (the cyclic and indicator features were kept unchanged), giving 23, 33, 53, 88 and 144 features. Coefficients were solved by least squares. Table 4 shows the results across degrees.

Table 4. Polynomial regression progression across degrees on traffic data.

Degree	Features	Condition number	Train MSE	Test MSE	Test RMSE	Test R^2
1 (OLS)	23	122	1,182,480	1,173,326	1,083.2	0.6969

Degree	Features	Condition number	Train MSE	Test MSE	Test RMSE	Test R^2
2	33	4,509	1,178,655	1,166,996	1,080.3	0.6986
3	53	1.0×10^{20}	1,178,157	1,166,866	1,080.2	0.6986
4	88	9.1×10^{23}	1,175,760	1,164,935	1,079.3	0.6991
5	144	1.8×10^{27}	3,930,451	3,833,350	1,957.9	0.0099

3.4 Underfitting versus overfitting

- *Underfitting.* A constant model (training mean) and a 4-feature continuous-only linear model fail to explain traffic periodicity (test R^2 = 0.000 and 0.027; MSE ≈ 3.8×10^6), showing high bias because calendar seasonality is omitted.
- *Overfitting in a controlled 300-sample experiment.* On the full dataset (N = 38,563), degrees 2–4 do not show variance overfitting because samples greatly exceed parameters. When restricted to a controlled N = 300 training subset, classical overfitting appears: training MSE decreases from 3.62×10^6 (degree 1) to 3.05×10^6 (degree 4), while test MSE rises to 5.39×10^7 at degree 4 (14 times the degree-1 baseline), illustrating severe variance.

• Degree 5 solver artefact. For degree 5 on continuous features, the design-matrix condition number reaches 3.2×1028 (far above double-precision machine epsilon ~1016). Scikit-learn's solver gives test R2 0.0099 (test MSE 3.87×106), while a stable SVD least-squares fit (numpy.linalg.lstsq with an explicit intercept) gives train R2 0.6910 and test R2 0.6851 (test MSE 1,219,247). This collapse is a solver artefact, not statistical variance overfitting. Degree 4 remains optimal.

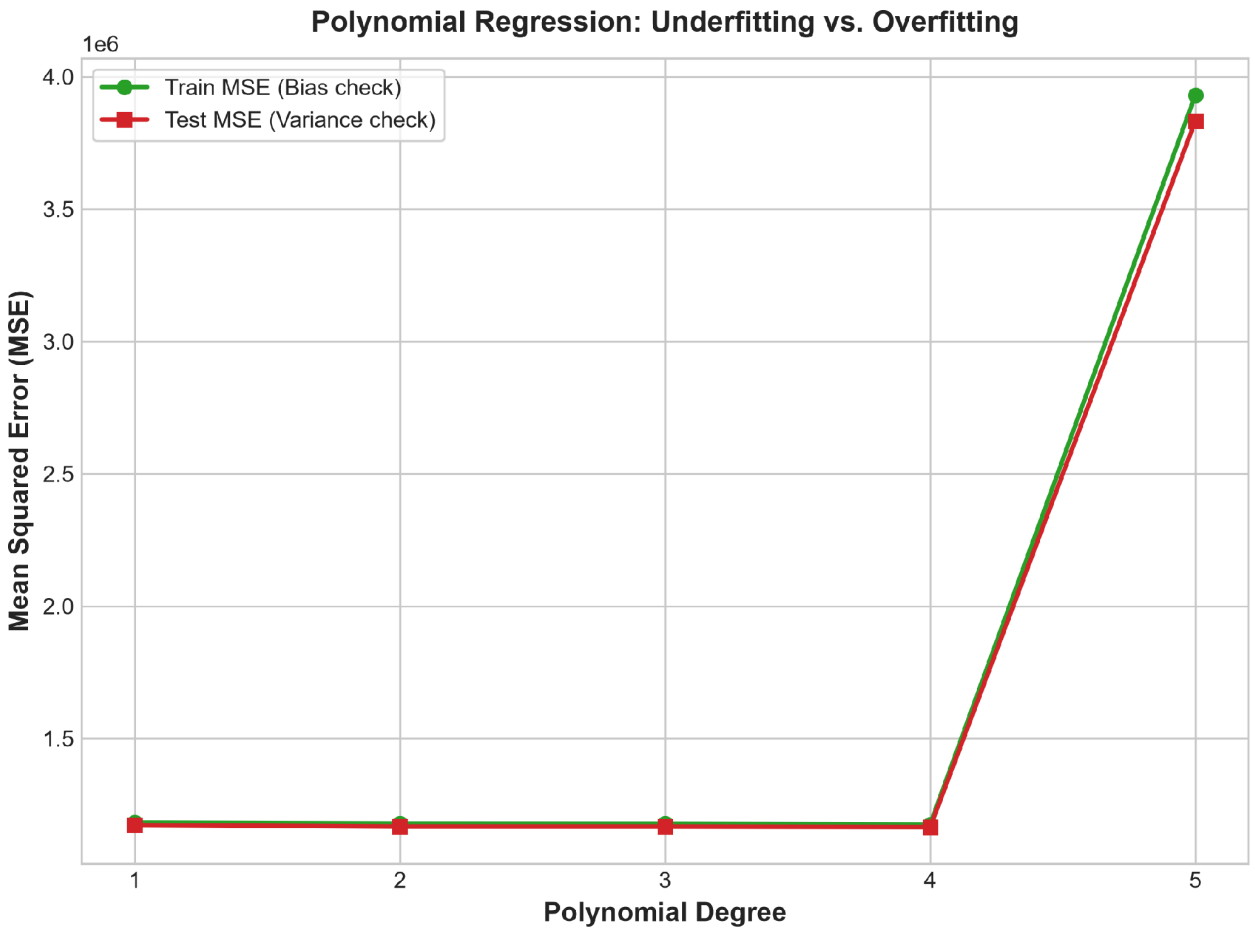


Figure 8. Training and test MSE against polynomial degree on the full training set.



Figure 9. Actual against predicted test traffic volume (polynomial regression, degree 4).

3.5 Ridge and lasso regularization

The regularization strength α was selected by 5-fold time-series cross-validation (forward-chaining splits that prevent look-ahead bias). Ridge was evaluated for α in [0.001, 1000] and selected α = 100 (CV MSE 1,196,747). Lasso was evaluated for α in [0.0001, 10] and selected α = 10 (CV MSE 1,195,765). At α = 10, lasso set 13 of the 23 coefficients to zero (all 11 weather indicators, the holiday flag and sin(month)), retaining the core cyclic time drivers and temperature. Because unregularized OLS and the regularized models perform within 0.1% of each other on test data, the 38,563 training observations provide sufficient constraint against multicollinearity.

Table 5. Consolidated regression benchmark on the held-out test set (9,641 records).

Regression model	Train MSE	Train RMSE	Train R^2	Test MSE	Test RMSE	Test R^2
Linear (batch GD)	1,183,099	1,087.7	0.7017	1,171,125	1,082.2	0.6975

Regression model	Train MSE	Train RMSE	Train R^2	Test MSE	Test RMSE	Test R^2
Polynomial deg 1 (OLS)	1,182,480	1,087.4	0.7019	1,173,326	1,083.2	0.6969
Polynomial deg 2	1,178,655	1,085.7	0.7028	1,166,996	1,080.3	0.6986
Polynomial deg 3	1,178,157	1,085.4	0.7030	1,166,866	1,080.2	0.6986
Polynomial deg 4	1,175,760	1,084.3	0.7036	1,164,935	1,079.3	0.6991
Polynomial deg 5	3,930,451	1,982.5	0.0091	3,833,350	1,957.9	0.0099
Ridge ($\alpha = 100$)	1,182,764	1,087.5	0.7018	1,172,175	1,082.7	0.6972
Lasso ($\alpha = 10$)	1,187,423	1,089.7	0.7006	1,176,769	1,084.8	0.6961

4. Part 3A: K-Means Clustering from Scratch (Gas Turbine Telemetry)

4.1 Dataset and algorithm

The gas-turbine dataset contains hourly sensor measurements collected between 2011 and 2015 from an industrial gas turbine in northwestern Turkey. After removing 7 duplicate rows, 36,726 clean observations of 11 continuous features remain: ambient temperature (AT), ambient pressure (AP), ambient humidity (AH), air filter difference pressure (AFDP), gas turbine exhaust pressure (GTEP), turbine inlet temperature (TIT), turbine after temperature (TAT), turbine energy yield (TEY), compressor discharge pressure (CDP), carbon monoxide (CO) and nitrogen oxides (NOx). All 11 features were standardized to zero mean and unit variance so that measurement units do not bias the Euclidean distance.

Lloyd's algorithm was implemented from scratch in NumPy. To avoid poor local minima, 10 random restarts were run for every K in {2, 3, 4, 5, 6}, iterating until the maximum centroid displacement fell below 10^{-4} or 300 iterations were reached. The run with the smallest within-cluster sum of squares was kept: $\mathrm{WCSS} = \Sigma_k \Sigma_{x_\{i} \in C_k\} \|x_i - \mu_k\|^2$.

Table 6. K-means clustering results for K in {2, ..., 6} (10 restarts per K).

K	Best WCSS	Reduction (%)	Mean WCSS	Worst WCSS	Silhouette	Iterations
1 (baseline)	403,986	–	–	–	–	–

K	Best WCSS	Reduction (%)	Mean WCSS	Worst WCSS	Silhouette	Iterations
2	276,942	31.45%	276,942	276,942	0.327	49
3	213,958	22.74%	213,958	213,958	0.286	28
4	180,981	15.41%	182,993	191,042	0.255	49
5	157,444	13.01%	161,254	165,171	0.274	35
6	142,087	9.75%	147,748	153,206	0.253	28

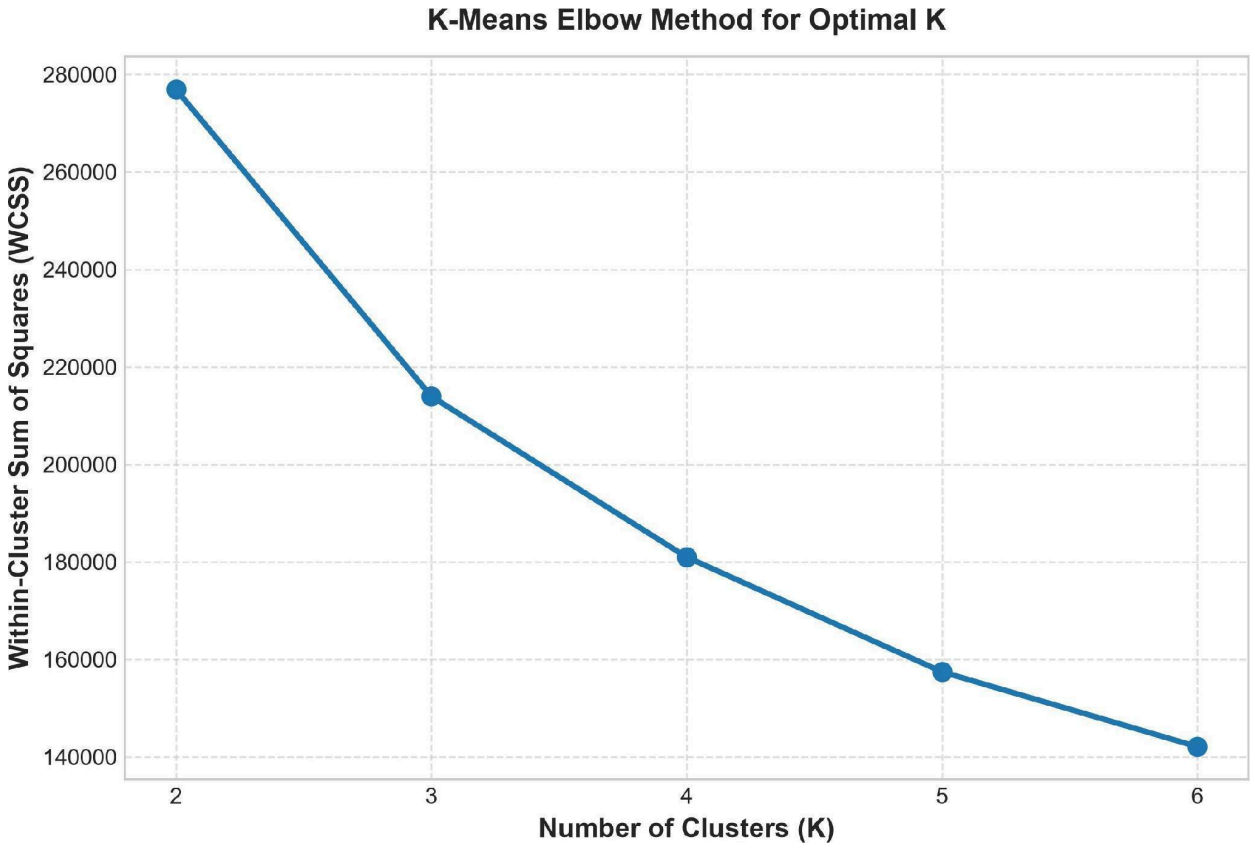


Figure 10. K-means elbow curve (WCSS) for K in {2, 3, 4, 5, 6}.

4.2 Cluster profiling and operational load interpretation

The WCSS curve bends continuously without an abrupt kink, while the silhouette score (computed on an 8,000-record sample for computational tractability) peaks at K = 2 (0.327). K = 3 (WCSS 213,958, silhouette 0.286) is selected as the most physically meaningful operational partition. Table 7 shows that the K = 3 centroids are ordered along turbine load intensity.

- Cluster 0 (low load, 8,435 records): lowest energy yield (TEY 112.2 MWh), compressor pressure (CDP 10.62 bar) and turbine inlet temperature (TIT 1,054.1 °C).

- Cluster 1 (mid load, 19,203 records): intermediate operating state (TEY 132.8 MWh, CDP 11.96 bar, TIT 1,084.8 °C), representing standard steady-state generation.
- Cluster 2 (peak load, 9,088 records): highest output (TEY 154.7 MWh, CDP 13.61 bar, TIT 1,099.7 °C).

Table 7. Profile of the K = 3 clusters (means in original physical units).

Cluster	Records	Mean TEY (MWh)	Mean CDP (bar)	Mean TIT (°C)	Mean AT (°C)	Operational regime
Cluster 0	8,435	112.2	10.62	1,054.1	16.1	Low load / off-peak
Cluster 1	19,203	132.8	11.96	1,084.8	18.4	Base load / steady operation
Cluster 2	9,088	154.7	13.61	1,099.7	17.6	Peak load / high dispatch

5. Part 3B: Principal Component Analysis from Scratch

5.1 Method

PCA was implemented from scratch with NumPy matrix operations and no high-level library wrappers. For the standardized matrix X (36,726 × 11) with zero column means, the sample covariance matrix $C = X^TX / (N - 1)$ (11 × 11) was formed. Its eigenvalues λ_j and orthogonal eigenvectors v_j were computed with a symmetric eigensolver (np.linalg.eigh) and sorted in descending order. The explained variance ratio is $EVR_j = \lambda_j / \Sigma\lambda$. Eigenvector orthonormality was verified to machine precision (largest deviation of V^TV from the identity 7.2×10^{-16}). The eigenvalues sum to 11.0003 rather than exactly 11, because the data were scaled with divisor N while the covariance uses N − 1; this does not affect the ratios.

Table 8. Eigenvalues, explained variance ratio and cumulative variance for the 11 components.

Component	Eigenvalue (λ)	Explained variance (%)	Cumulative variance (%)
PC1	5.2801	48.00	48.00
PC2	2.1894	19.90	67.90 (2D projection)
PC3	1.0390	9.45	77.35 (3D projection)

Component	Eigenvalue (λ)	Explained variance (%)	Cumulative variance (%)
PC4	0.8398	7.63	84.98
PC5	0.6887	6.26	91.24 (above 90% threshold)
PC6	0.4828	4.39	95.63
PC7	0.2597	2.36	97.99
PC8	0.1762	1.60	99.59
PC9	0.0343	0.31	99.91
PC10	0.0083	0.08	99.98
PC11	0.0021	0.02	100.00

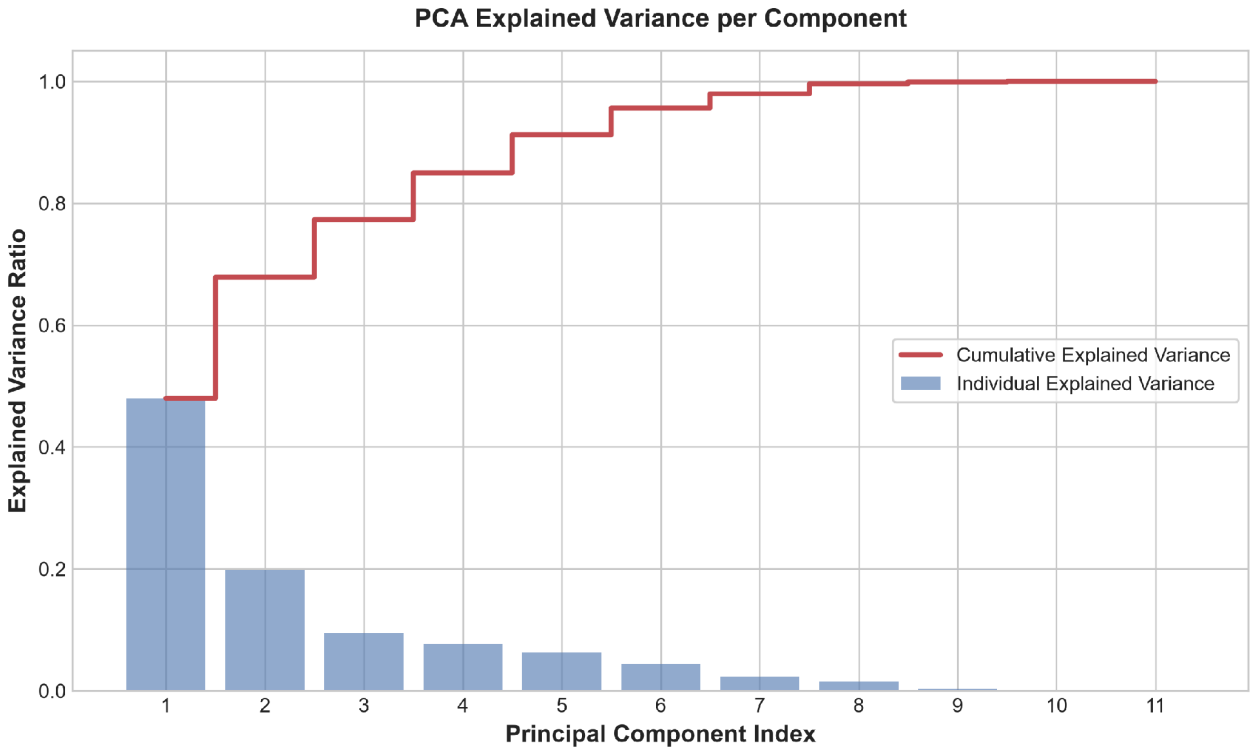


Figure 11. Individual and cumulative explained variance ratio (scree plot).

5.2 Eigenvector loadings

Table 9 lists the loadings of the first three components; the sign of an eigenvector is arbitrary. PC1 is dominated by CDP (−0.427), GTEP (−0.421), TEY (−0.421) and TIT (−0.405), with opposing weights on TAT (+0.279) and CO (+0.281). PC1 therefore acts as a turbine-load axis.

PC2 is dominated by ambient variables (AT +0.605, AP −0.379, AH −0.294) and NOx emissions (−0.458). PC3 reflects humidity (+0.773) and pressure (−0.428).

Table 9. Eigenvector loadings of the first three principal components.

Feature	PC1	PC2	PC3	Dominant physical association
AT	−0.051	+0.605	−0.194	Ambient temperature (weather)
AP	−0.025	−0.379	−0.428	Ambient pressure (weather)
AH	+0.108	−0.294	+0.773	Ambient humidity (weather)
AFDP	−0.339	+0.073	+0.035	Air filter differential pressure
GTEP	−0.421	−0.058	−0.046	Exhaust pressure (turbine load)
TIT	−0.405	+0.072	+0.101	Inlet temperature (combustion)
TAT	+0.279	+0.344	+0.181	After temperature (exhaust heat)
TEY	−0.421	−0.131	+0.045	Turbine energy yield (power output)
CDP	−0.427	−0.084	−0.018	Compressor discharge (turbine load)
CO	+0.281	−0.193	−0.352	Carbon monoxide (emissions)
NOx	+0.110	−0.458	−0.097	Nitrogen oxides (emissions)

5.3 Dimensionality reduction and 2D projection

The 3-component projection has shape 36,733 × 3 (retaining 77.35% cumulative variance). Projecting the 11-dimensional telemetry onto the first two principal components, Z = X [v1 v2], preserves 67.90% of the total variance. Figures 12 and 13 show the resulting 2D projection.

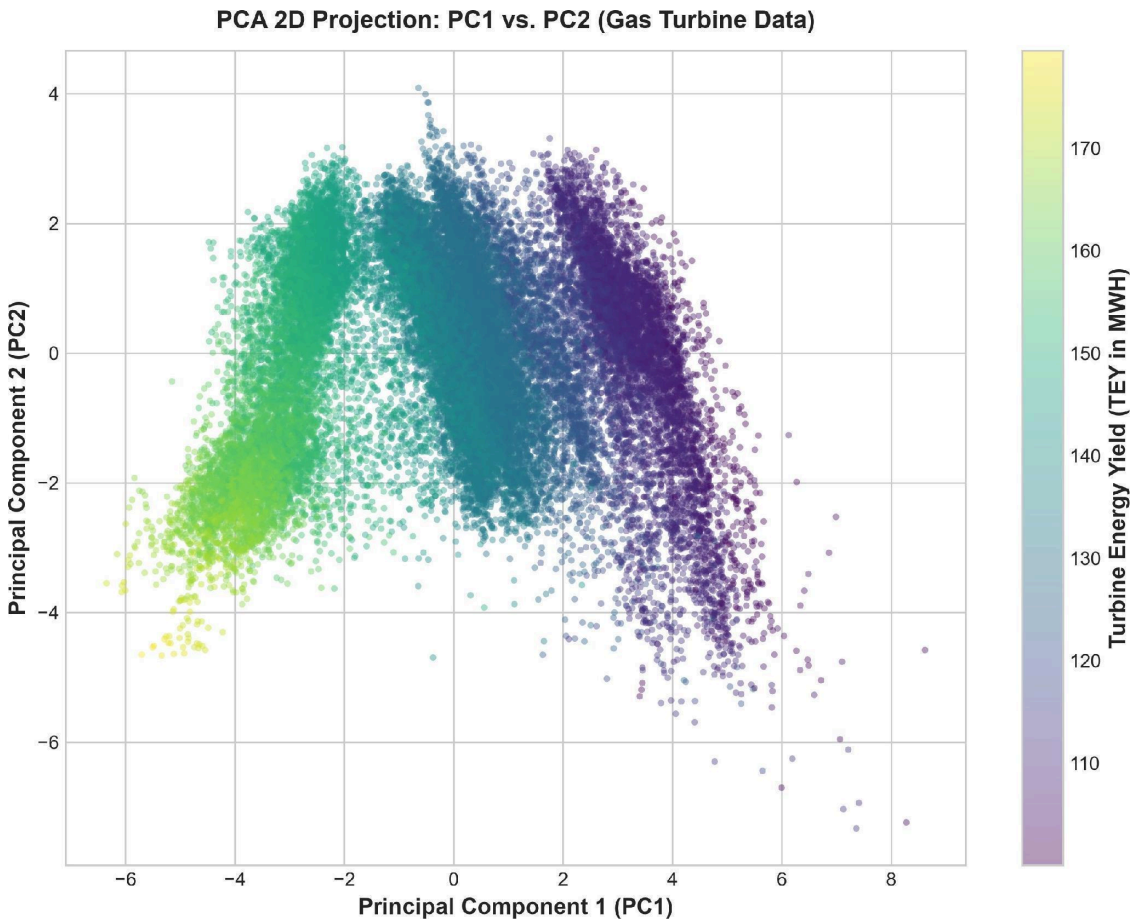


Figure 12. 2D PCA projection shaded by turbine energy yield (TEY).

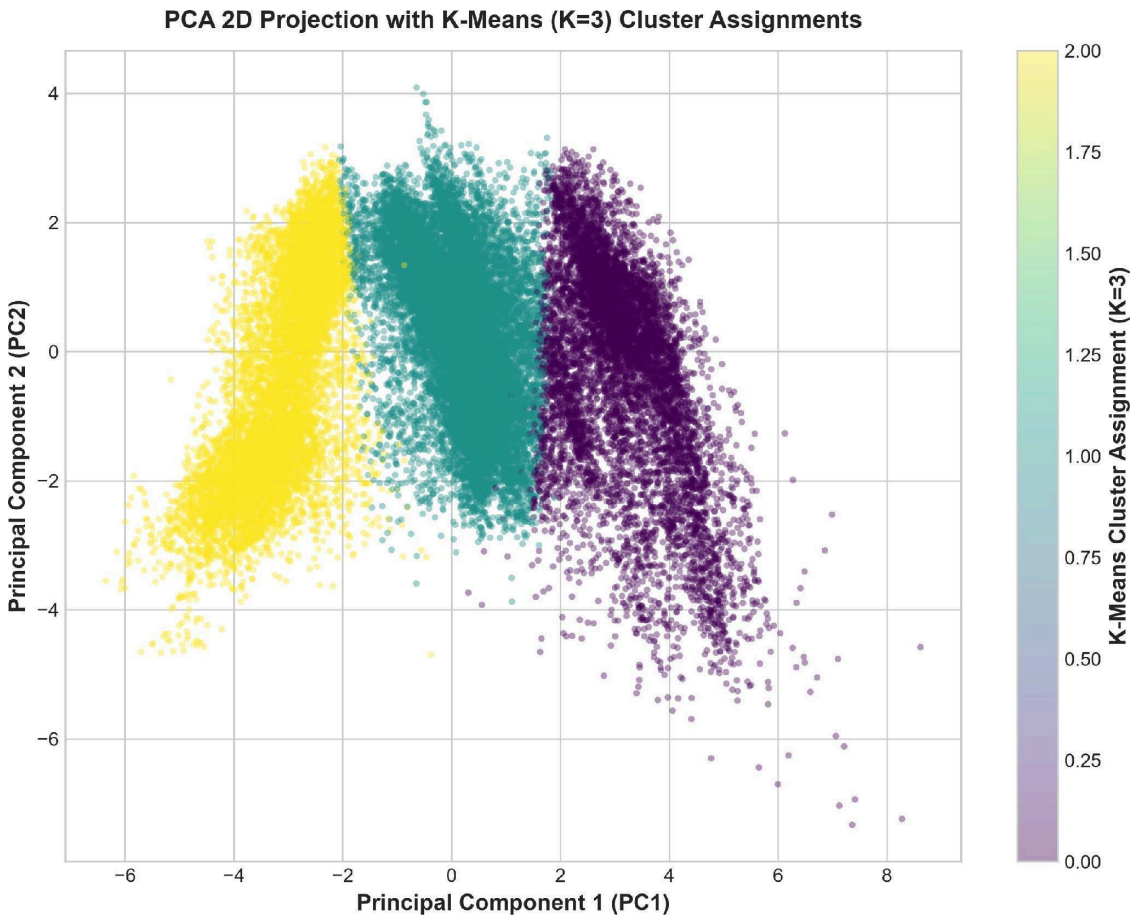


Figure 13. 2D PCA projection with K-means (K = 3) cluster assignments.

The 2D projection shows a continuous, curved cloud rather than separated islands. In Figure 12, turbine energy yield shows a smooth gradient along PC1 from right to left (reflecting the negative loadings on TEY and CDP). In Figure 13, the K = 3 clusters partition the cloud into three vertical slices along PC1, consistent with K-means separating the low, medium and high load regimes.

6. Limitations and Methodological Discussion

- **Image feature representation.** The traditional ML models used 1,764-dimensional HOG features, which capture spatial edges but lack hierarchical context, giving lower accuracy than the deep CNN. SVM training time was substantial because of single-threaded LibSVM kernel computations.
- **Polynomial conditioning.** Degree 5 regression is severely ill-conditioned and numerically unstable, with a condition number of 1.8×10^{27}.
- **Time-series generalization.** In Part 2 the chronological split respects temporal order; however, test performance also reflects distribution shifts across years.

- **Cluster selection.** The K-means elbow is gradual; K = 3 is an operationally useful partition rather than a mathematical global optimum.
- **CNN bonus.** The 5-mark CNN bonus was implemented and benchmarked on a GPU.

7. Reproducibility and Project Artifacts

All code, configuration, metrics and figures are organized for one-command reproducibility:

- python experiments/run_classification.py : full classification pipeline (stages 1–7), generating all metrics and comparison plots.
- python experiments/run_regression.py : Part 2 linear, polynomial, ridge and lasso regression experiments.
- python experiments/run_clustering.py : Part 3 K-means and PCA from-scratch implementations. All generated outputs reside in outputs/ and are synchronized to reports/figures/ and reports/results/.

8. References

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3. Kaya, H., Tüfekci, P., and Uzun, E. (2019). Gas Turbine CO and NOx Emission Data Set. UCI Machine Learning Repository (ID 551).